

suspended inside, jointly developed by German companies Cairos Technologies and Adidas athletic clothing and shoe company. The technology uses a network of receivers around the field designed to track the ball's precise position in real time, including exactly when it fully passed the goal line. That information is relayed in less than a second to a watch-like device worn by the referee.

Hawk-Eye and GoalControl systems

The system by the UK company Hawk-Eye has had very successful trials and is the most used technology at the moment. Hawk-Eye has already been used to make calls in tennis and cricket. The football version has been extensively trailed and has performed very well in the test environment and the field.

The system uses a network of high-speed video cameras to track a ball's position at a given time via triangulation; each taking footage at 600 frames a second. Hawk-Eye is able to give a definitive decision on whether the ball has fully crossed the line, and relay this information in the form of an audible beep to the central referee within half a second. As Premier League referees use headsets, signal is easily sent to them. Other methods such as through a watch are used.

The system's software can also predict the future path of a ball; it is often used like this in cricket to determine whether the ball would have hit the wicket had it not hit the batsman first, and the predicted path can be drawn onscreen. Because it can create this kind



Fig. 3: Frank Lampard's shot clearly crosses the line



Fig. 4: Adidas miCoach smartball



Fig. 5: Camera locations for Hawk-Eye

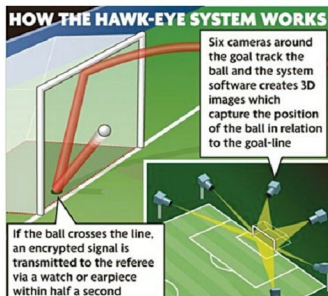


Fig. 6: How Hawk-Eye works

of visual display, it is expected to be popular among viewers of televised games.

As used in tennis, Hawk-Eye has a margin of error of just 3.6mm, better than the 3cm required by football's governing body, FIFA. However, it needs to be able to see at least a quarter of the ball to work, which is not always possible if there are multiple legs in the way. In 2013, English Football Association announced that Hawk-Eye would be used in 2013-14 Premier League season, where implementation turned out to be a success.

GoalControl is a German company that has developed a goal line technology for association football. The GoalControl system features 14 high-speed cameras, which are mounted around the stadium, seven directed to each of the goals. These cameras are used to detect if the ball has crossed the goal line or not. The cameras are capable of taking 500 pictures per seconds and the ball's movement within 5mm.

Camera data is sent to the central image processing centre located within the stadium using fibre-optic cables, where a virtual representation of the ball is output to confirm the goal. The referee is equipped with a watch displaying the camera's data, which vibrates and displays a signal upon a goal. The goal is transmitted wirelessly. The system was used in 2014 World Cup in Brazil where it was the first World Cup to feature goal line technology.

Goal Ref system

Goal Ref goal-detection system has been developed by